Questions and Dataset

Examine the dataset carefully and determine which attribute would you select as the Target variable (y) to create the most impactful predictive model for the specified domain. Provide a convincing justification for your choice.

Based on the target variable you chose in Question 1, would you apply K-Nearest Neighbors (KNN) or Linear Regression to build your model? Explain your choice by discussing the relationship between the features and the target (Draw a simple to support your explanation).

Describe your strategy for the training and testing of the model. Which specific evaluation metrics would you use, explain your choice.

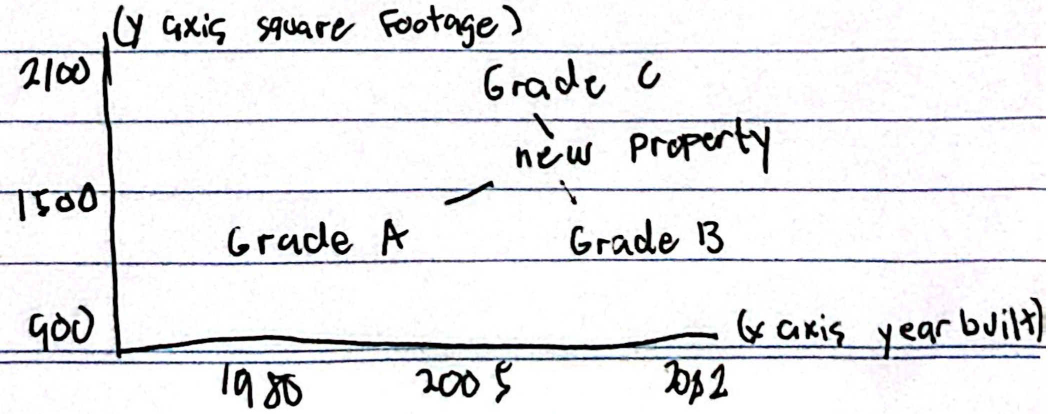
Property Developer

Property ID	Square Footage	Distance to City Center (Km)	Year Built
501	1200	2.5	1995
502	850	10.2	2015
503	2100	5.0	1980
504	950	1.8	2022
505	1550	7.4	2005

Energy Rating	Parking Available
Grade A	yes
Grade C	no
Grade B	yes
Grade A	no
Grade B	yes

1) I think choosing Energy Rating as my target variable will give my predictive model the impactfulness it needs. I chose Energy Rating as my target variable because I think its the most reliable based on the given dataset. I think its reliable because the dataset is entitled Property Developer and as a developer for properties, looking for which property has the better energy rating, I will be able to determine which property has the better energy sustainability. Why do I want to focus on energy sustainability because focusing on which property sustains energy better, will be a higher value property or will be the "eye cather" property. ?

2) Since I chose Energy Rating as my target variable, I will be applying K-Nearest Neighbors (KNN) to build my model. I chose knn because it uses classification models that predicts categorical or discrete outputs. Since my target variable is Energy Rating this will be put into classification because it consists of specific grades (Grade A, B, C). Using knn will be in good use because it will operate on a logical assumption being properties with similar physical traits will likely have similar energy rating. Using the euclidean distance formula, it will calculate the distance between the new and old properties in the dataset. Then it will look at the 'K' closest neighbors and determine the new property.



3) My strategy for testing and training this model is by having an aggressive and necessary data preparation phase. For me, if I just give this raw dataset to my model since its using knn it will be pratically useless. Since the Square footage nembers are much bigger than the Distance to City Center the calculation for the formula will be too much to handle. So using data normalization will be a necessary first stop to rescale each feature to a common range.